HTTP://WWW.MAKAUT.COM

(b) Prove that the average case time complexity of quick sort is $O(n \log n)$. Clearly state the reason behind the design of the recurrence relation you use to compute the complexity. 7

8. (a) Given the weight vector (2, 3, 5, 7, 1, 4, 1) and profit vector (10, 5, 15, 7, 6, 18, 3) and a knapsack of capacity 20. Find at least three feasible solutions including the optimal one for the knapsack problem of seven objects. 10

(b) Explain P class, NP class, NP hard and NP complete class and describe their relationships. 5

9. (a) Describe the Dijkstra's algorithm for single source shortest path problem. 8

(b) Derive the time complexity of Dijkstra's algorithm. 4

(c) Compare between DFS and BFS algorithms. 3

10. (a) Compare dynamic programming and greedy approach. 3

(b) Write down the algorithm for travelling salesman problem using dynamic programming and greedy approach. 4

(c) Explain which approach among these two for travelling salesman problem is better and why? 4

11. Write short notes on any *three* of the following. 3×5

(a) Strassen's Matrix multiplication

(b) Ford-Fulkerson's algorithm

(c) Master theorem

(d) Floyd-Warshall algorithm

(e) 8–queen's problem

CS/B.Tech/IT/Odd/Sem-5th/IT-501/2014-15

IT-501

Design and Analysis of Algorithm

Time Allotted: 3 Hours Full Marks: 70

The questions are of equal value
The figures in the margin indicate full marks
Candidates are required to give their answers in their own words as far as practicable

GROUP A
(Multiple Choice Type Questions)

1. Answer all questions. 10×1 = 10

(i) Time complexity of the following recurrence $T(n) = 25T(n/5) + n$ is

(A) $O(n \log n)$ (B) $\theta(n^2)$
(C) $\theta(\log n)$ (D) $O(n^3)$

(ii) Which edge removal makes the graph disconnected?

(A) pendant vertex (B) bridge
(C) articulation point (D) colored vertex

(iii) Lower bound of time complexity of any comparison based sorting algorithm is

(A) $O(\log n)$ (B) $O(n)$
(C) $O(n \log n)$ (D) $O(n^2)$

(iv) For two sorted lists each of size n, the maximum comparisons in merge is

(A) $2n$ (B) $n + 1$
(C) $2n - 1$ (D) $2n + 1$

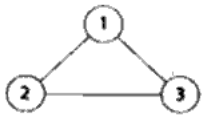
(v) Space complexity of quick sort is

(A) $O(\log n)$ (B) $O(n)$
(C) $O(n \log n)$ (D) $O(n^2)$

HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM

(vi)



The chromatic number of the above graph is

(A) 1 (B) 2
(C) 3 (D) 4

(vii) _____ is used in BFS

(A) stack (B) queue
(C) link list (D) none of these

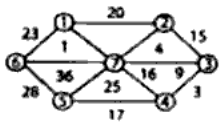
(viii) Ω-notation (Big Omega) is Asymptotically _____ bound.

(A) tight (B) upper
(C) lower (D) loose

(ix) Time complexity of insertion sort is

(A) linear (B) quadratic
(C) cubic (D) exponential

(x)



Total cost of the spanning tree for the above graph using Kruskal's algorithm is

(A) 57 (B) 55
(C) 64 (D) 52

HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM

CS/B.Tech/IT/Odd/Sem-5th/IT-501/2014-15

GROUP B
(Short Answer Type Questions)

Answer any *three* questions. 3×5 = 15

2. Define θ-notation. 2+1.5×2

$$L(n) = \begin{cases} 0 & \text{for } n = 1 \\ L(\lfloor n/2 \rfloor) & \text{for } n > 1 \end{cases}$$

where $\lfloor k \rfloor$ denote the floor of k i.e. greatest integer which does not exceeds.
(a) Find L(25)
(b) What will be the complexity of L?

3. Mr. B.C Dull obtained astonishing result by using the 'Self-evident' formula $O(f(n)) - O(f(n)) = 0$. What was his mistake and what should the right hand side of his formula have been? 2+3
Prove that if $f(n) = a_m n^m + a_{m-1} n^{m-1} + \ldots + a_1 n + a_0$ then $f(n) = O(n^m)$.

4. Considering all the cases calculate the complexity of Binary search. 5

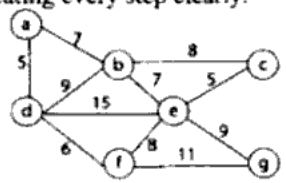
5. Prove that the time complexity of Tower of Hanoi problem is exponential. 5

6. Discuss Job Sequencing with Deadlines with an example. 5

GROUP C
(Long Answer Type Questions)

Answer any *three* questions. 3×15 = 45

7. (a) Find the minimum cost spanning tree of the following graph using Prim's algorithm indicating every step clearly. 8



HTTP://WWW.MAKAUT.COM

HTTP://WWW.MAKAUT.COM